

## Oral Cavity in Disease (OBC302)

### Lectures 15 & 16: Diseases of Pulp & Diseases of Periapical Region

**Intended Learning Outcome: After reading this chapter, the student should be able to:**

1. Classify the different diseases of pulp.
2. Describe the etiopathogenesis and clinicopathological features of pulpal diseases.
3. Classify the different diseases of the Periapical region.
4. Describe the etiopathogenesis and clinicopathological features of periapical diseases

#### **Outline**

##### **I. Introduction**

##### **II. Pulpal Disease**

1. Normal Pulp
2. Reversible Pulpitis
3. Irreversible Pulpitis (Symptomatic & Asymptomatic)
4. Pulp Necrosis
5. Previously Treated
6. Previously Initiated Therapy

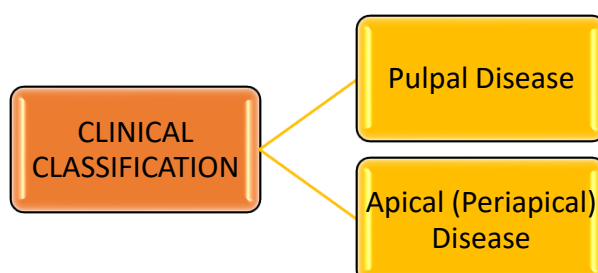
##### **III. Apical (Periapical) Disease**

1. Normal Apical Tissues
2. Symptomatic Apical Periodontitis
3. Asymptomatic Apical Periodontitis
4. Acute Apical Abscess
5. Chronic Apical Abscess

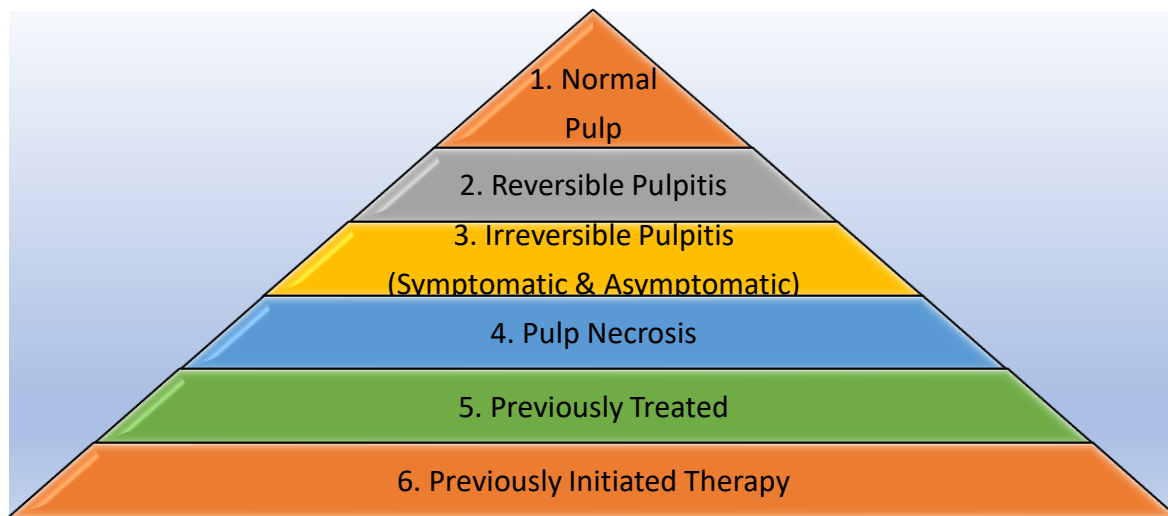
#### **I. Introduction**

#### **CLINICAL CLASSIFICATION OF PULPAL AND PERIAPICAL DISEASES**

The following terminology and classifications are based on those suggested by the American Association of Endodontists in 2012.



## I. Pulpal Disease



### Pulpitis

- This is a clinical and histologic term denoting inflammation of the dental pulp, clinically described as reversible or irreversible and histologically described as acute, chronic, or hyperplastic.

#### 1. Normal Pulp

##### Def:

is a clinical diagnostic category in which the pulp is symptom-free and normally responsive to pulp testing.

##### Clinically

- Although the pulp may not be histologically normal, a “clinically” normal pulp results in a mild or transient response to thermal cold testing, lasting no more than one to two seconds after the stimulus is removed.
- One cannot arrive at a probable diagnosis without comparing the tooth in question with adjacent and contralateral teeth. It is best to test the adjacent teeth and contralateral teeth first so that the patient is familiar with the experience of a normal response to cold.



## 2. Reversible Pulpitis

### Def

Inflammation should *resolve*, and the pulp return to normal.

### Signs and Symptoms

- ▶ When the pulp within the tooth is *irritated* so that the *stimulation* is *uncomfortable* to the patient but *reverses quickly* after irritation.

### Causative factors

- ▶ *Caries, exposed dentin, recent dental treatment, and defective restorations.*

### Treatment

- ▶ Conservative *removal* of the irritant will *resolve* the symptoms.



## 3. A. Symptomatic Irreversible Pulpitis

Def. Clinical diagnosis based on subjective and objective findings indicating that the *vital inflamed* pulp is *incapable of healing*.

### Signs and Symptoms

- ▶ *include intermittent or spontaneous pain.*
- ▶ Rapid exposure to *dramatic temperature* changes (especially to cold stimuli) will *elicit* heightened and *prolonged* episodes of *pain* even after the thermal stimulus has been removed.
- ▶ The pain in these cases may be *sharp or dull, localized, diffuse, or referred.*
- ▶ Deep restorations, caries, pulp exposure, or any other direct or indirect insult to the pulp, recently or historically, may be present.
- ▶ Patients who *present* with symptomatic anterior teeth for which there are *no obvious* etiologic factors should also be questioned regarding past *general anesthesia, endotracheal intubation* procedures, or history of *orthodontic* treatment.



### Radiographically

- ▶ Typically, there are *minimal or no changes* in the radiographic appearance of the periradicular bone.

- ▶ With advanced irreversible pulpitis, a *thickening* of the *periodontal ligament* may become apparent on the *radiograph*, and there may be some evidence of pulpal irritation by virtue of *extensive* pulp chamber or root canal space *calcification*.



### Treatment

- ▶ Typically, when symptomatic irreversible pulpitis remains untreated, the pulp will eventually become *necrotic*.

### **3.B. Asymptomatic Irreversible Pulpitis**

Def. Vital *inflamed* pulp is *incapable* of *healing*.

### Signs and symptoms

- ▶ The patient, however, *does not complain* of any symptoms.
- ▶ On occasion, *deep caries* will not produce any symptoms, even though clinically or radiographically, the caries may extend well into the *pulp*. Left *untreated*, the tooth may become *symptomatic*, or the *pulp* will become *necrotic*.



### Treatment Endodontic treatment

should be performed **as soon as** possible so that symptomatic irreversible pulpitis or necrosis does not develop and causes the patient severe pain and distress.

#### 4. Pulp Necrosis

Def indicating the *death* of the dental pulp.

##### Signs and symptoms

- ▶ The pulp is usually *nonresponsive* to pulp testing.
- ▶ The pulpal *blood* supply is nonexistent, and the pulpal nerves are *nonfunctional*.
- ▶ The tooth may become *hypersensitive* to heat (possibly related to remnants of fluid or gases in the *pulp* canal space *expanding* and extending into the *periapical tissues*.) Even to the warmth of the oral cavity, and is often relieved by applications of *cold*.
- ▶ The tooth will typically become *asymptomatic* until such time when there is an extension of the disease process into the *periradicular tissues*, the tooth may become *symptomatic* to *percussion* or exhibit *spontaneous* pain.
- ▶ Pulpal necrosis may be *partial* or *complete*, and it may not involve all of the canals in a *multirooted tooth*; the tooth may present with *confusing symptoms*. Pulp testing over one root may give no response, whereas, over *another root*, it may give a *positive* response.
- ▶ Pulp necrosis, in the *absence* of *restorations*, caries, or luxation injuries, is likely caused by a *longitudinal fracture* extending from the occlusal surface and into the *pulp*.



##### Radiographic

- ▶ *Thickening* of the *periodontal* ligament space to the appearance of a *periapical radiolucent* lesion.

## 5. Previously Treated

Def. Indicating that the tooth has been *endodontically* treated and the canals are *obtured* with *various filling* materials other than intracanal medicaments.

- ▶ The tooth *may* or may *not* present with signs or symptoms but will require additional *nonsurgical* or *surgical endodontic* procedures to *retain* the tooth. In most such situations, there will *no longer* be any *vital* or *necrotic pulp* tissue present to respond to pulp testing procedures.



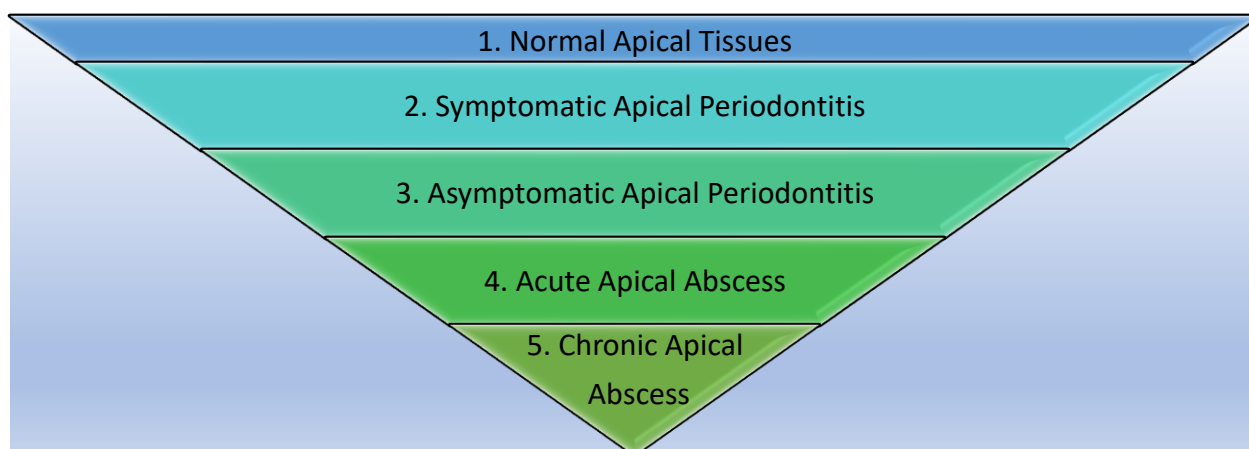
## 6. Previously Initiated Therapy

Def. indicating that the tooth has been *previously* treated by partial endodontic therapy (e.g., *pulpotomy*, *pulpectomy*).

- ▶ In most instances, partial endodontic therapy was performed as an *emergency procedure* for symptomatic or asymptomatic irreversible pulpitis cases.
- ▶ In other situations, these procedures may have been performed as *part* of *vital pulp* therapy procedures, *traumatic* tooth injuries, *apexification*, or apexogenesis therapy.
- ▶ At the time these cases present for root canal therapy, it would *not be possible* to make an *accurate pulpal diagnosis* because all, or *part*, of the *pulp tissue* has *already been removed*.



### III. Apical (Periapical) Disease



#### 1. Normal Apical

##### Def

Tissues are not sensitive to percussion or palpation testing and radiographically, the lamina dura surrounding the root is intact and the periodontal ligament space is uniform.

- As with pulp testing, comparative testing for percussion and palpation should always begin with normal teeth as a baseline for the patient.

#### 2. Symptomatic Apical Periodontitis

Def. An *inflammation*, usually of the *apical* periodontium, produces clinical symptoms including a *painful* response to *biting* or *percussion* or *palpation*.

##### Pulp test

- This tooth *may* or *may not* respond to *pulp vitality* tests.

##### Radiographic

- It *might* or *might not* be associated with an apical *radiolucent* area.
- A radiograph or image of the tooth will typically exhibit *at least* a *widened periodontal ligament* space.



### 3. Asymptomatic Apical Periodontitis

Def. *Inflammation and destruction of apical periodontium that is of pulpal origin, appears as an apical radiolucent area, and does not produce clinical symptoms.*

#### Symptoms

- Generally *not sensitive* to biting pressure but may "*feel different*" to the patient on percussion.

#### Pulp test

- *Does not* usually respond to pulp *vitality* tests,

#### Radiographic

- Will exhibit an *apical radiolucency*.



### 4. Acute Apical Abscess

Def *Inflammatory reaction to pulpal infection and necrosis characterized by rapid onset, spontaneous pain, tenderness of the tooth to pressure, pus formation, and swelling of associated tissues.*

#### Symptoms

- *Acutely painful* to biting pressure, percussion, and palpation.
- *Swelling* will be present *intraorally* and the *facial* tissues adjacent to the tooth will almost always present.
- The patient will frequently be febrile, and the *cervical* and *submandibular lymph nodes* may exhibit *tenderness* to palpation.



#### Pulp test

- *Not respond* to any pulp vitality tests and will exhibit varying degrees of mobility.



## Radiographic

Image can exhibit *anything* from a *widened periodontal* ligament space to an apical *radiolucency*.

## 5. Chronic Apical Abscess

Def Inflammatory reaction to pulpal infection and necrosis characterized by *gradual* onset, little or no *discomfort*, and the *intermittent discharge* of *pus* through an associated sinus tract.

### Symptoms

- ▶ Will *not present* with clinical symptoms.
- ▶ Usually, the tooth is not sensitive to biting pressure but can "*feel different*" to the patient on percussion. This entity is distinguished from asymptomatic apical periodontitis because it will exhibit *intermittent drainage through an associated sinus tract*.



### Pulp test

- ▶ The tooth will *not* respond to pulp vitality tests,

### Radiographic

Apical *radiolucency*.



### References

Pathways of the Pulp 12th Edition. Kenneth M Hargreaves, Stephen Cohen, Elsevier Publishing Co. PART I. The Core Science of Endodontics chapter 1. Diagnosis.

**Good Luck**

**Amr Elnaghy**